

PHAYAO, Thailand

WMO: 483100

Lat: 19.1931N

Lon: 99.8836E

Elev: 399

StdP: 96.62

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.5	12.9	8.7	7.3	21.5	10.3	8.2	19.9	4.1	26.1	3.4	25.8	0.0	320	0.270

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.5	37.0	23.4	35.8	23.6	34.7	23.7	27.0	32.1	26.6	31.6	26.2	31.0	1.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	22.0	29.5	25.2	21.4	29.0	25.0	21.0	28.7	87.9	32.1	86.0	31.8	84.3	31.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.4	3.6	3.1	9.0	38.5	2.0	1.1	7.5	39.3	6.3	39.9	5.1	40.5	3.7	41.3		
(5)			8.5	28.0	2.1	0.8	7.0	28.6	5.8	29.1	4.7	29.5	3.2	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	21.9	24.1	27.0	29.2	28.5	28.3	27.7	27.4	27.3	26.4	24.2	21.9
	DBStd	3.04	2.04	2.03	2.28	2.16	2.08	1.28	1.32	1.12	1.13	1.48	2.06	2.50
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	7	2	0	1	0	0	0	0	0	0	0	0	4
	CDD10.0	5894	369	393	526	575	574	549	548	538	518	509	427	368
	CDD18.3	2859	113	160	268	325	316	299	289	279	268	251	177	113
	CDH23.3	27368	845	1488	2901	3838	3448	3215	2846	2569	2347	1926	1208	737
CDH26.7	11114	252	657	1508	2054	1580	1256	989	819	764	650	396	188	
Wind	WSAvg	0.5	0.3	0.5	0.6	0.7	0.7	0.9	0.8	0.6	0.3	0.2	0.2	0.3
Precipitation	PrecAvg	1175	12	9	36	81	177	104	158	214	212	120	39	13
	PrecMax	1545	76	82	184	204	309	212	284	385	362	229	159	102
	PrecMin	826	0	0	0	9	14	27	55	93	87	28	0	0
	PrecStd	210	20	18	45	47	67	43	60	76	67	58	41	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	34.8	37.2	38.9	38.2	35.2	34.4	33.4	33.2	33.3	32.7	31.4
		MCWB	20.5	20.3	21.6	23.7	25.5	25.8	25.8	26.0	26.0	25.7	24.6	23.1
	2%	DB	30.5	33.5	36.1	37.5	36.1	33.9	33.1	32.3	32.3	32.1	31.3	30.0
		MCWB	20.6	20.4	21.6	23.6	25.4	25.8	25.6	25.9	25.8	25.3	24.0	22.2
	5%	DB	29.3	32.3	35.0	36.3	34.6	32.9	32.0	31.4	31.5	31.2	30.3	28.8
		MCWB	20.5	20.5	21.6	23.8	25.5	25.6	25.5	25.7	25.6	25.0	23.5	21.8
	10%	DB	28.1	31.0	33.7	35.0	33.1	31.9	31.1	30.5	30.6	30.2	29.1	27.5
		MCWB	20.2	20.4	21.7	23.8	25.4	25.5	25.4	25.5	25.4	24.7	22.9	21.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.8	23.9	25.5	26.6	27.4	27.3	27.2	27.1	27.2	26.7	25.8	24.7
		MCDB	28.4	30.4	32.4	34.5	33.9	32.4	31.7	31.4	31.4	31.3	30.7	29.7
	2%	WB	22.5	23.0	24.3	25.8	26.7	26.7	26.5	26.5	26.6	26.1	25.0	23.6
		MCDB	27.3	29.5	31.5	33.5	33.0	31.7	30.9	30.7	30.7	30.5	29.6	27.9
	5%	WB	21.7	22.2	23.6	25.3	26.2	26.2	26.0	26.1	26.1	25.6	24.3	22.7
		MCDB	26.6	28.6	30.8	32.7	32.2	31.1	30.4	30.2	30.2	29.7	28.6	26.7
	10%	WB	21.0	21.5	23.0	24.7	25.7	25.8	25.6	25.7	25.7	25.1	23.7	21.9
		MCDB	26.0	27.9	30.2	31.9	31.3	30.4	29.8	29.5	29.4	28.9	27.6	25.9
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.4	14.0	11.5	8.7	7.2	6.7	6.6	7.1	8.3	10.6	12.2
		MCDBR	15.5	17.0	15.8	13.2	11.1	8.7	8.3	7.9	8.4	9.5	11.6	12.9
		MCWBR	6.4	5.9	4.7	3.6	3.4	2.7	2.8	2.8	3.0	3.4	4.7	5.6
	5% WB	MCDBR	12.6	14.1	12.9	11.4	9.4	7.7	7.4	7.3	7.8	8.5	10.0	11.0
		MCWBR	5.9	5.6	4.7	3.9	3.5	2.9	2.9	3.0	3.1	3.4	4.3	5.2
Clear-Sky Solar Irradiance	taub	0.432	0.520	0.685	0.667	0.496	0.441	0.433	0.452	0.478	0.509	0.427	0.425	
	taud	2.198	1.962	1.604	1.702	2.190	2.375	2.391	2.317	2.224	2.091	2.285	2.246	
	Ebn at Noon	840	782	673	692	813	851	859	848	826	782	840	834	
	Edh at Noon	136	182	269	246	149	123	121	131	143	158	124	126	
All-Sky Solar Radiation	RadAvg	4.51	5.30	5.50	5.92	5.38	4.79	4.42	4.38	4.59	4.47	4.55	4.23	
	RadStd	0.25	0.24	0.32	0.36	0.47	0.33	0.32	0.26	0.19	0.22	0.32	0.26	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.31	N/A	N/A	N/A	N/A	N/A	N/A	+152	+150

Nomenclature: See separate page